

CSE422 LAB PROJECT REPORT

Predicting Customer Subscription to Bank Term Deposits Using Machine Learning

Binary Classification and Customer Clustering

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Section	13
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Course	CSE422

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1. Introduction

Bank marketing campaigns contact customers and offer products such as term deposits. The response is not the same for every customer, so a prediction model can help analyse which campaign records are more likely to end in a subscription.

The aim of this project is to predict whether a customer subscribed to a term deposit after a marketing contact. The output column y has two values, yes and no, so this is a binary classification problem. Three supervised models are used: Logistic Regression, Gaussian Naive Bayes and a Neural Network. Their results are compared using accuracy, precision, recall, confusion matrices, ROC curves and AUC scores.

The project also treats the data as an unsupervised learning problem. K-means clustering is applied without using the target column so that customer groups can be explored from age and call duration.

Project objective: Build and compare classification models for term-deposit subscription, then use K-means to show customer clusters.

2. Dataset Description

2.1 Dataset Overview

The supplied Bank Marketing dataset contains 41,188 data points and 21 columns. Twenty columns are input features and y is the output feature. The output contains two classes, yes and no, which confirms that the task is classification rather than regression.

Data points	Input features	Output feature	Problem type
41,188	20	y (yes/no)	Binary classification

2.2 Feature Types

There are 10 quantitative input features and 10 categorical input features. Categorical inputs must be encoded because the selected scikit-learn models require numerical input.

Feature type	Features
Quantitative (10)	age, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed
Categorical (10)	job, marital, education, default, housing, loan, contact, month, day_of_week, poutcome

2.3 Initial Data Quality Check

No standard null values were found. However, six categorical columns contain the label unknown and 12 rows are exact duplicates. The largest unknown count is in default, with 8,597 records. Education has 1,731 unknown records, while job, marital, housing and loan contain smaller numbers.

2.4 Exploratory Data Analysis

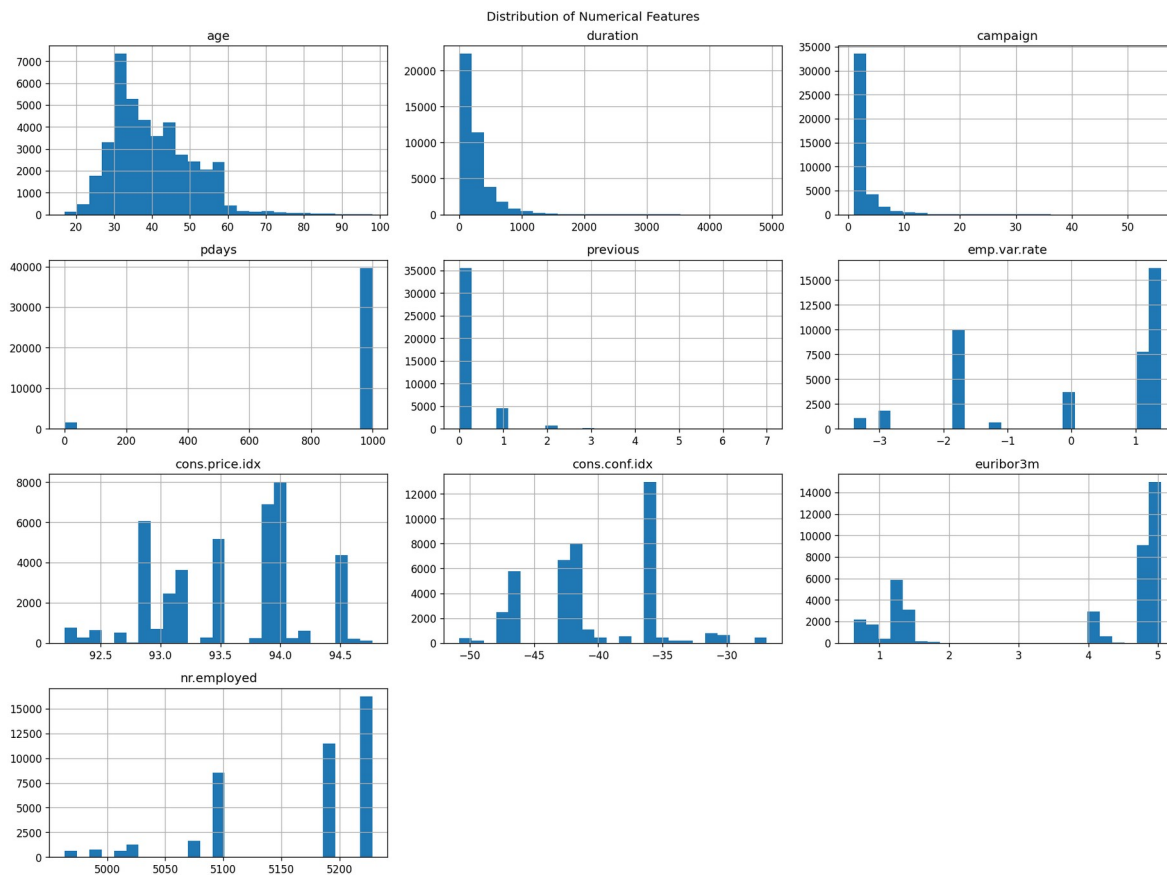


Figure 1. Histograms of the numerical features

Most customers are between about 30 and 50 years old. Call duration is strongly right-skewed because most calls are short and a small number are very long. Campaign is also right-skewed, showing that most customers were contacted only a few times. The value 999 dominates pdays and means that most customers were not previously contacted.

2.5 Outlier Check

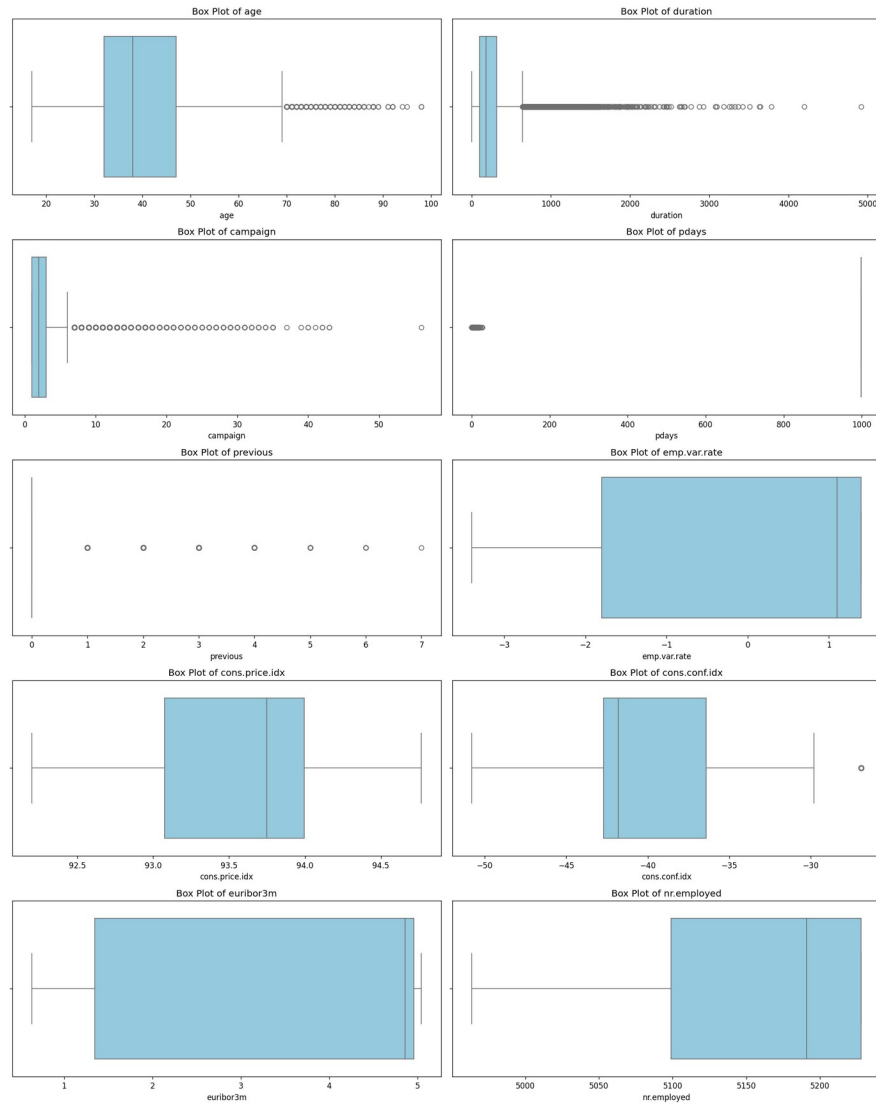


Figure 2. Box plots of the numerical features

The box plots show high values in duration, campaign, previous and age. These observations were not removed automatically because they can represent real campaign behaviour. StandardScaler was later used so features with larger numerical ranges would not dominate the models.

2.6 Imbalanced Output Feature



Figure 3. Distribution of the output classes

The classes are not equal. The no class contains 36,548 records (88.73%), while the yes class contains 4,640 records (11.27%). Because of this imbalance, accuracy alone is not enough. Precision, recall, confusion matrices and AUC are also used.

2.7 Correlation Test

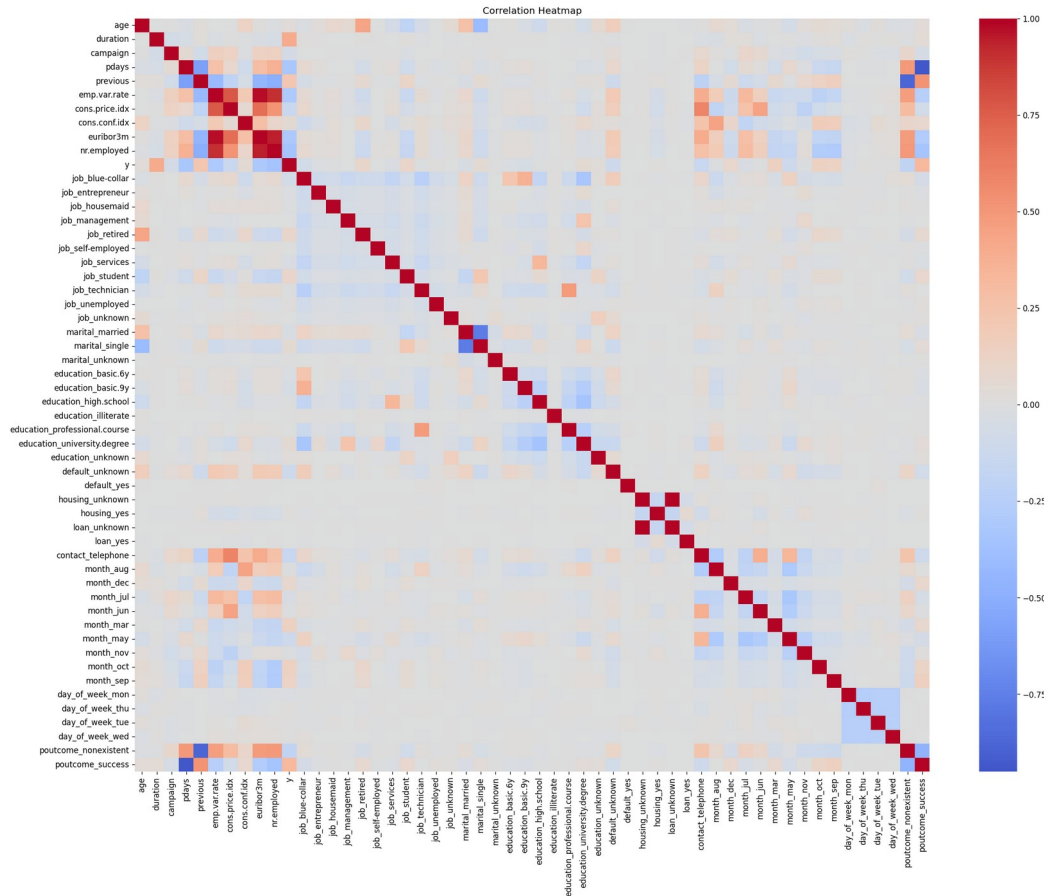


Figure 4. Correlation heatmap after encoding

Duration has the strongest positive correlation with y at about 0.405. Successful previous campaign outcome also has a positive relationship. $nr.employed$, $pdays$, $euribor3m$ and $emp.var.rate$ have notable negative correlations. These values show association, not proof that one feature directly causes subscription.

3. Dataset Pre-processing

The guideline asks each data problem to be discussed together with its solution. The preprocessing was therefore completed in the following order.

3.1 Problem: Missing and Duplicate Records

Check: The `isnull()` check returned zero for every column, so no numerical or categorical value required imputation. The `duplicated()` check found 12 repeated rows.

Solution: No row was removed for missing values. The 12 exact duplicate rows were removed using `drop_duplicates()`, reducing the working dataset from 41,188 to 41,176 rows.

3.2 Problem: Unknown Categorical Labels

Check: The word unknown appears in job, marital, education, default, housing and loan. It is a category label, so pandas does not count it as a null value.

Solution: Unknown was kept as its own category. Removing every affected row would discard many records, especially because default alone contains 8,597 unknown values.

3.3 Problem: Categorical Values

Solution: The target was mapped as no = 0 and yes = 1. The ten categorical input columns were converted using one-hot encoding with `drop_first=True`. This produced 53 numerical input columns and avoided assigning an artificial order to category names.

3.4 Problem: Different Feature Scales

Solution: `StandardScaler` was fitted only on the training data. The fitted scaler was then used to transform both training and test inputs. This prevents information from the test set from affecting preprocessing.

Duration decision: Duration was retained because this project predicts the result after a marketing call. For a model that selects customers before calling, duration should be removed because it is not known in advance.

4. Dataset Splitting

After preprocessing, the input data X contained 53 columns and the output y contained the binary target. The data was divided with an 80% training set and a 20% test set. A stratified split was used so the same class ratio was maintained in both parts. `random_state=42` was used so the split could be reproduced.

Subset	Rows	Input columns	Yes class
Training	32,940	53	11.27%
Testing	8,236	53	11.27%

4.1 Why Stratification Was Required

Only about 11% of the records belong to the yes class. A normal random split could produce slightly different class proportions in training and testing. Stratification protects the evaluation by keeping the imbalance consistent.

4.2 Neural Network Validation

The `MLPClassifier` used `early_stopping=True` and `validation_fraction=0.20`. It therefore held out 20% of the training data internally for validation. The separate test set remained untouched until final evaluation.

Final split: The models learned from 32,940 training rows and were evaluated on 8,236 unseen test rows. The test set was not balanced or oversampled.

5. Model Training and Testing

5.1 Logistic Regression

Logistic Regression was trained on the scaled training features with `max_iter=1000`. It estimates the probability of the yes class and provides a simple linear baseline for binary classification.

5.2 Gaussian Naive Bayes

Gaussian Naive Bayes was trained on the same scaled data. It calculates class probabilities using Bayes' rule and assumes that the input features are conditionally independent with Gaussian distributions within each class.

5.3 Neural Network

The neural network was implemented with `MLPClassifier`. It used two hidden layers with 64 and 32 neurons, ReLU activation, Adam optimisation, a learning rate of 0.001, batch size 128 and a maximum of 100 iterations. Early stopping monitored an internal validation set. Training stopped after 9 iterations when validation performance no longer improved enough.

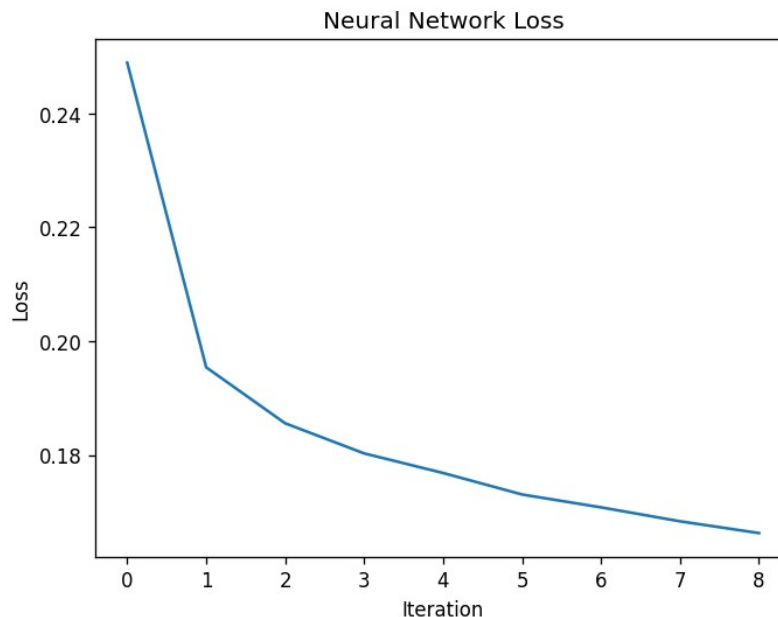


Figure 5. Neural network training loss

The loss decreased during training, which shows that the network was learning from the training data. Early stopping limited further iterations and helped reduce unnecessary training and overfitting.

5.4 K-means Clustering

For the unsupervised part, the target y was excluded. K-means was applied to age and duration so that the clusters could be shown clearly in two dimensions, following the clustering method used in the lab. These two inputs were scaled separately before clustering.

Choosing the Number of Clusters

The elbow method calculated the sum of squared errors for K values from 1 to 9. The reduction is large up to $K=3$ and becomes more gradual after that point, so three clusters were selected.

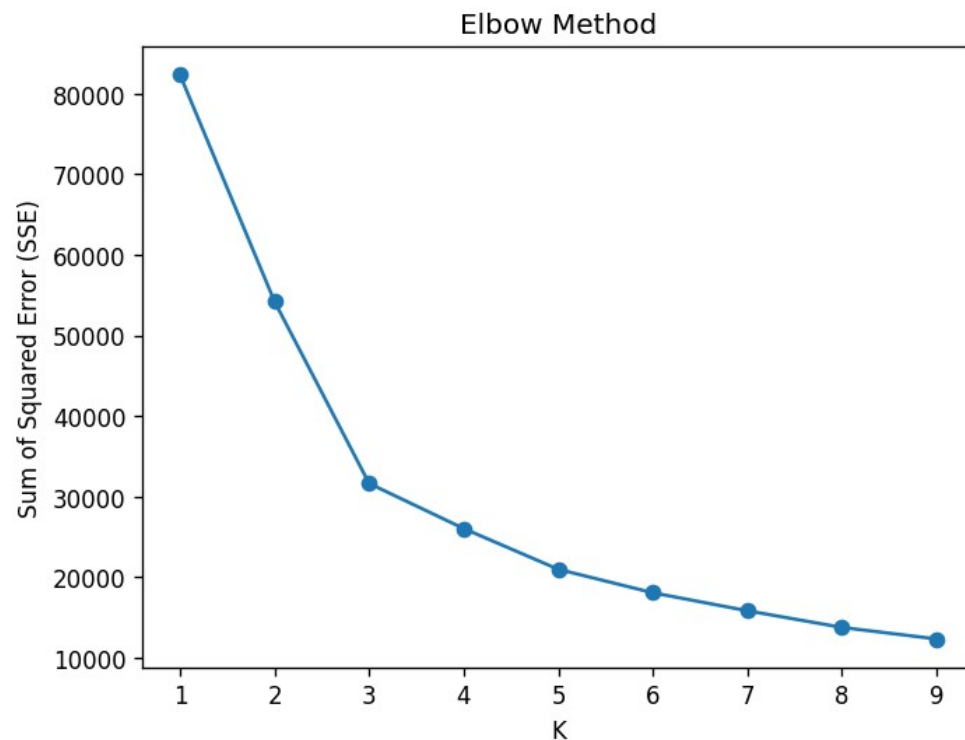


Figure 6. Elbow method for selecting K

The SSE decreased from 82,352 at $K=1$ to 54,208 at $K=2$ and 31,641 at $K=3$. Later reductions were smaller, supporting $K=3$ as a simple and interpretable choice.

5.5 K-means Cluster Results

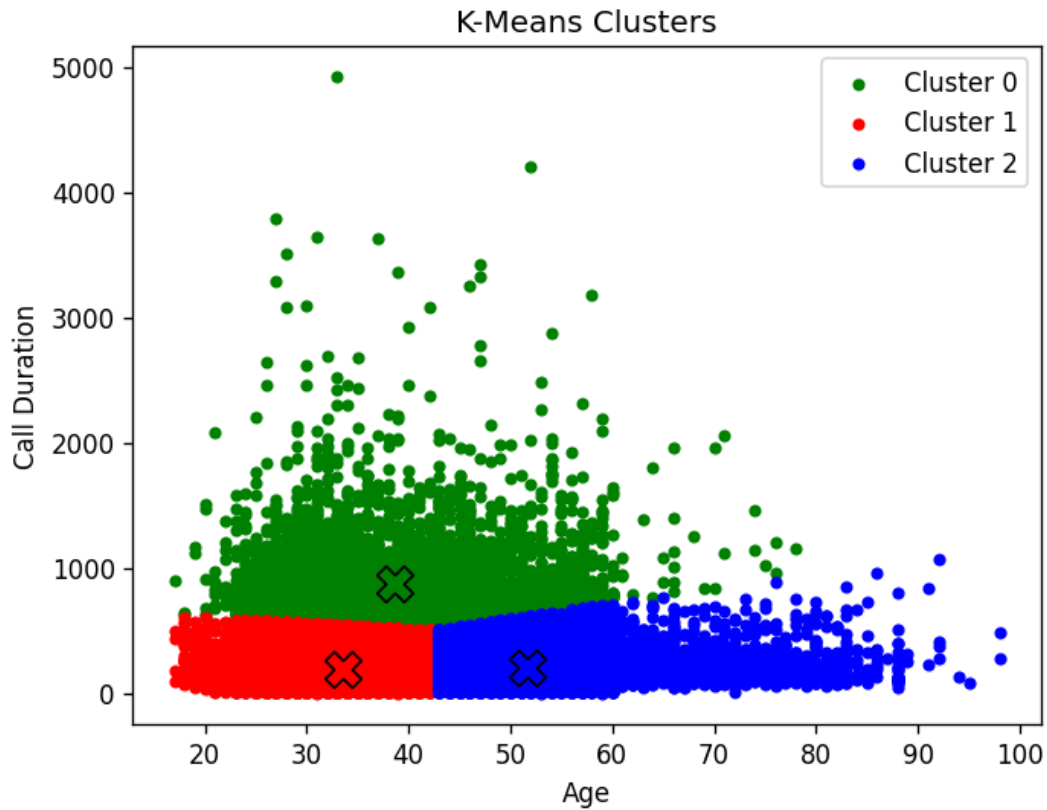


Figure 7. Customer clusters based on age and duration

Cluster	Records	Mean age	Mean duration	Yes rate
0	3,912	38.68	874.74 sec	45.73%
1	23,513	33.48	189.89 sec	7.19%
2	13,751	51.59	199.96 sec	8.44%

Cluster 0 contains customers with much longer calls and has the highest observed subscription rate. Cluster 1 mainly contains younger customers with short calls. Cluster 2 contains older customers with similarly short calls. The target was compared with the clusters only after K-means finished, so it did not influence cluster formation.

6. Model Selection and Comparison Analysis

All three supervised models were tested on the same 8,236 records. Accuracy measures overall correctness, precision measures how often a predicted yes is correct, recall measures how many actual yes cases were found, and AUC measures ranking performance across classification thresholds.

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	0.9093	0.6501	0.4224	0.9390
Gaussian Naive Bayes	0.7591	0.2965	0.8287	0.8508
Neural Network	0.9109	0.6359	0.4892	0.9392

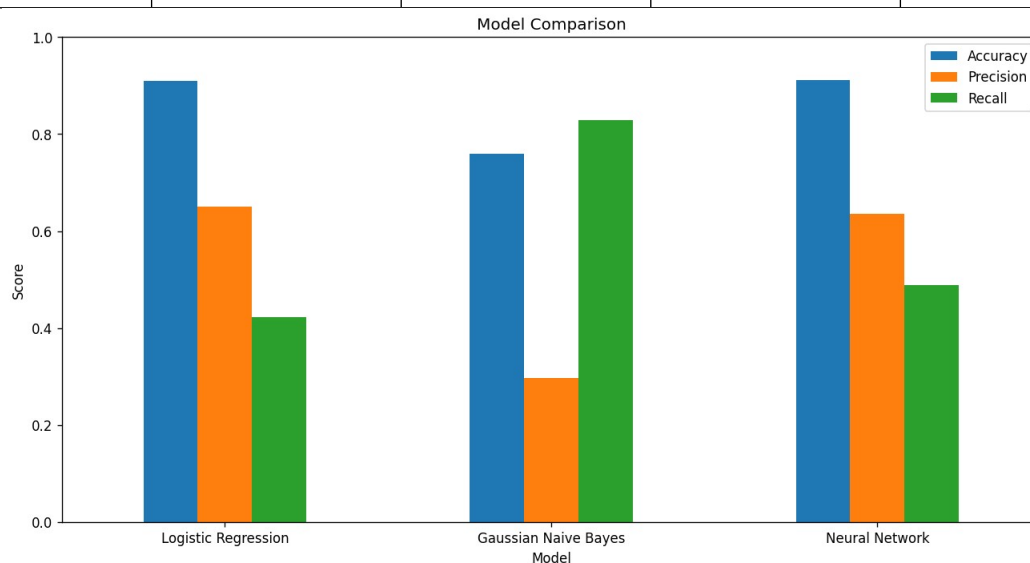


Figure 8. Accuracy, precision and recall comparison

The Neural Network achieved the highest accuracy and AUC, while Logistic Regression achieved the highest precision. Gaussian Naive Bayes achieved the highest recall, but its lower precision means it produced many false positive yes predictions.

6.1 Confusion Matrix Analysis

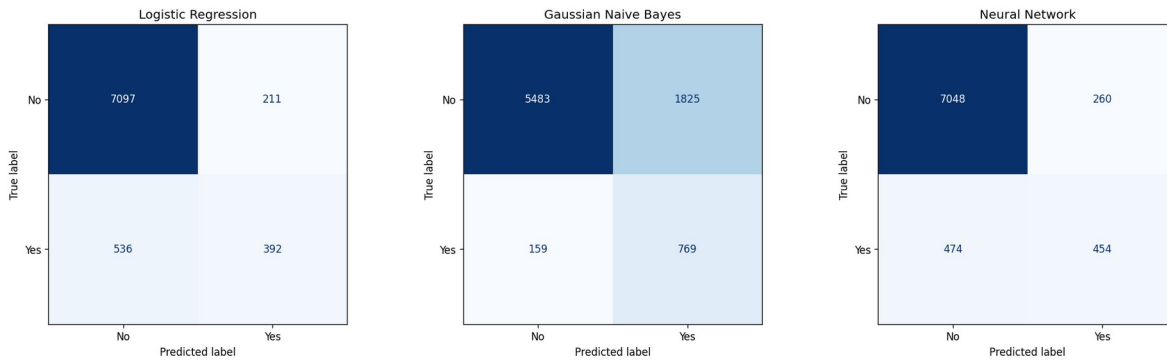


Figure 9. Confusion matrices for the three models

Logistic Regression correctly predicted 7,097 no cases and 392 yes cases. Gaussian Naive Bayes found 769 yes cases, but incorrectly labelled 1,825 no cases as yes. The Neural Network correctly predicted 7,048 no cases and 454 yes cases, giving a better balance than Naive Bayes.

6.2 ROC Curve and AUC Analysis

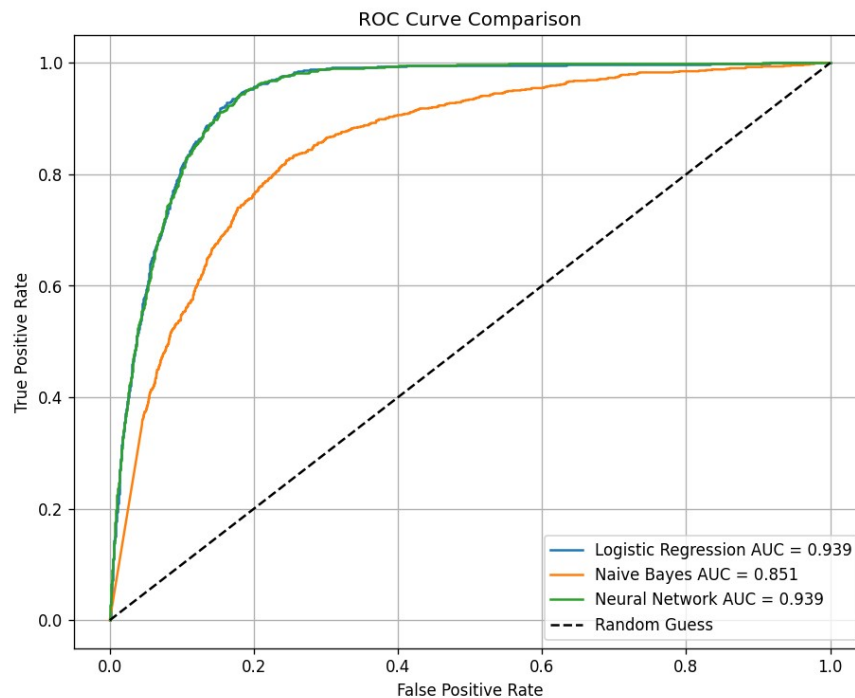


Figure 10. ROC curves and AUC scores

The Neural Network produced the highest AUC at 0.9392, followed very closely by Logistic Regression at 0.9390. Gaussian Naive Bayes had a lower AUC of 0.8508. Based on the full set of results, the Neural Network was selected as the best overall model, while Logistic Regression remained a strong and simpler alternative.

7. Conclusion

This project used the Bank Marketing dataset to predict whether a customer subscribed to a term deposit. The data contained both numerical and categorical features, no standard missing values, 12 duplicate rows and a strongly imbalanced target. The duplicates were removed, unknown labels were kept as valid categories, categorical inputs were one-hot encoded and numerical inputs were standardised using training data only.

The Neural Network gave the best overall result with 0.9109 accuracy, 0.4892 recall and 0.9392 AUC. Logistic Regression performed almost as well and had the highest precision of 0.6501. Gaussian Naive Bayes found the largest number of actual subscriptions, producing 0.8287 recall, but it also produced many false positives and therefore had much lower precision and accuracy.

The results are affected by the unequal class distribution. Since almost 89% of records are no, a model can achieve high accuracy while still missing many yes cases. This is why the final comparison used recall, precision, confusion matrices and AUC in addition to accuracy.

K-means formed three understandable groups from age and call duration. The cluster with long calls had a much higher observed subscription rate. This result agrees with the correlation analysis, where duration had the strongest positive relationship with the target.

Challenges and Final Comments

- The target classes were highly imbalanced, so model quality could not be judged from accuracy alone.
- Several columns used unknown as a category instead of a normal null value, so these labels had to be checked separately.
- Duration is highly useful for prediction but is only known after the call, which limits its use in a pre-call targeting system.
- K-means required scaled numerical features and a clear choice of K before the clusters could be interpreted.

Final selection: The Neural Network is the best overall model for this experiment because it has the highest accuracy and AUC with a better recall than Logistic Regression. Logistic Regression is the simpler alternative with slightly higher precision.